Ujjwal Basumatary

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Education

Indian Institute of Science, Bangalore Nov 2020 – Apr 2025
Bachelor and Master of Science, Major in Physics *(Physics: 9.6/10) CGPA: 9.3/10*

Publications and Preprints

- `arXiv:2410.22702` *Phys. Rev. D 111, L041306*
  Beyond Hawking evaporation of black holes formed by dark matter in compact stars, **Ujjwal Basumatary**, Nirmal Raj and Anupam Ray.
- `arXiv:2603.26090` *Under review at JHEP*
  Cosmological Correlators Using Tensor Networks, **Ujjwal Basumatary**, Aninda Sinha and Xinan Zhou.
- `arXiv:2512.02706` *Under review at TRiPS, Hindustan Book Agency/Springer*
  Lectures on Quantum Field Theory on a Quantum Computer, Aninda Sinha and **Ujjwal Basumatary**.
- *In preparation*
  Thermalization and entanglement in the Yukawa–SYK model, **Ujjwal Basumatary** and Sumilan Banerjee.

Employment

Lodha Theoretical Physics Institute, Mumbai Jul 2026 – Present
Research Assistant

Indian Institute of Science, Bangalore Aug 2025 – Jun 2026
Project Intern *Center for High Energy Physics*

Raman Research Institute, Bangalore Aug 2025 – Jun 2026
Project Intern, National Quantum Mission *Light and Matter Physics*

Indian Institute of Science, Bangalore Jun 2025 – Jul 2025
Project Associate - 1

Teaching Assistantships

HE381: Quantum Field Theory on a Quantum Computer *IISc, Bangalore*
Instructor: Prof. Aninda Sinha *CHEP*

- Conducted tutorials and designed/graded assignments, with demonstrations from basic quantum algorithms to Matrix Product States (MPS). Implementations: `qftonqc`.

Fellowships and Awards

S. N. Bhatt Memorial Fellowship May 2023 – Jul 2023
International Center for Theoretical Sciences (ICTS) *Bangalore*

Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship Nov 2020 – Apr 2025
Department of Science and Technology (DST) *India*

National Talent Search Exam (NTSE) Scholarship Apr 2018 – Mar 2020
National Council of Educational Research & Training (NCERT) *India*

Skills

- **Programming:** Julia, Python, Mathematica, C, C++ and Haskell.
- **Markup:** LaTeX, HTML and CSS.

Advanced Graduate Coursework

Non-eq. Many-Body Dynamics	10/10	AdS_3/CFT_2	9/10
Standard Model	10/10	Adv. Mathematical Physics	10/10
QFT - 2	9/10	Cosmology for Theorists	9/10
Statistical Field Theory	9/10	QFT - 1	10/10
General Relativity	10/10	Nuclear and Particle Physics	10/10